ComplAInSG

A Community Safety Network

An AI-Assisted Incident Reporting & Community Response Platform

Technical Documentation

Team Cortex | DLW 2026

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Architecture and Agentic Pipeline Visualisations

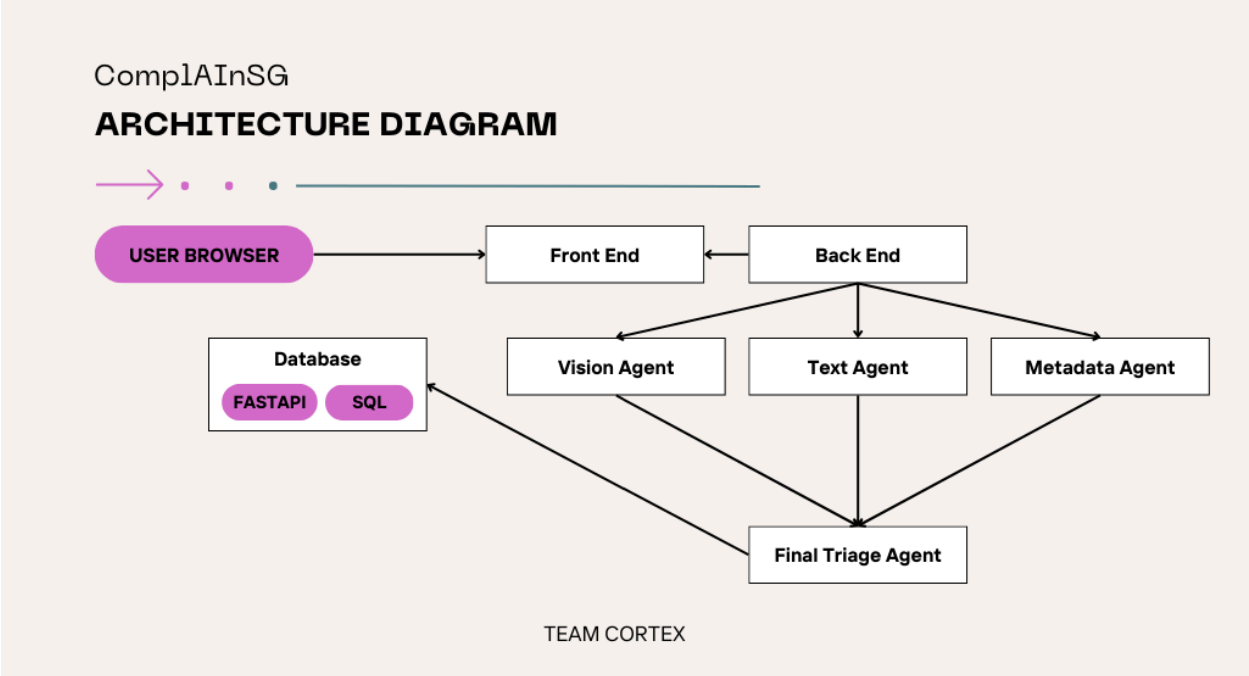

Figure 1. Diagram visualising our Application's Architecture.

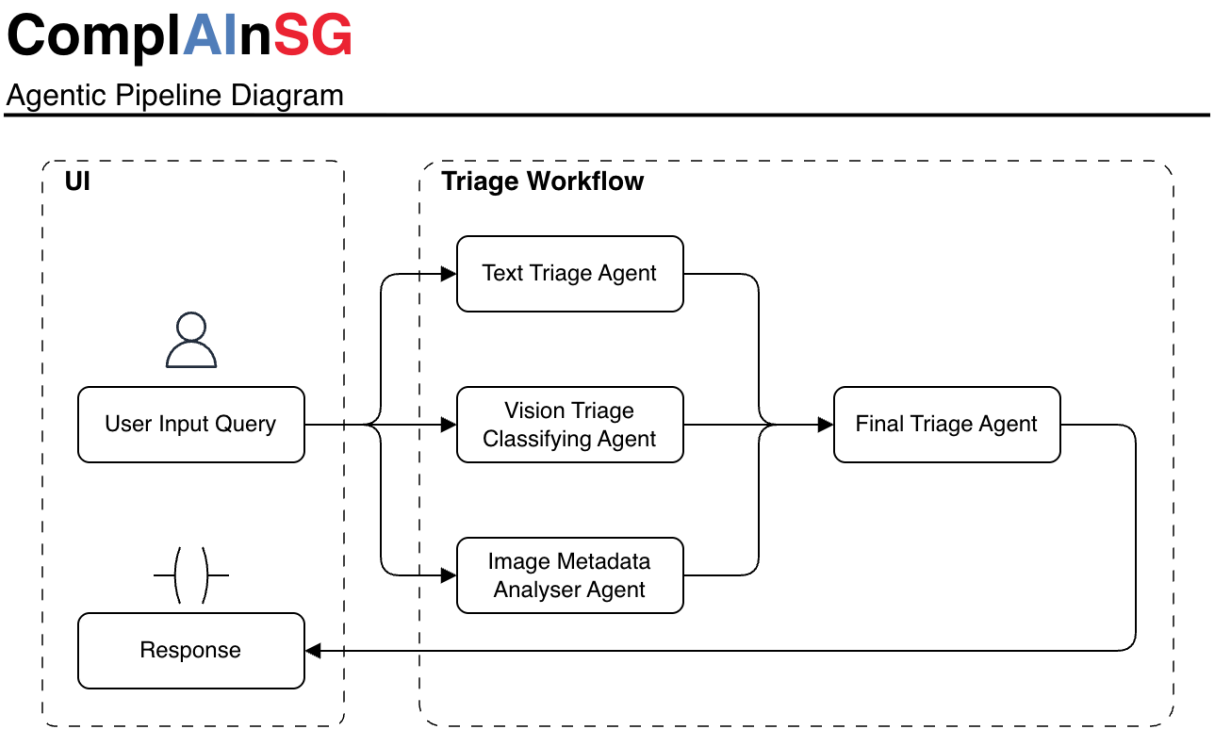

Figure 2. Diagram showcasing our Custom Agentic Pipeline for Incident Report Triaging.

1. Overview

ComplAInSG is an AI-powered mobile-first web app that transforms every smartphone into a community first-responder tool. It is built specifically to address a critical gap in public safety infrastructure, namely, the lag between when an incident occurs and when effective help arrives.

Singapore's police average response time is approximately 8 to 12 minutes for Priority 1 calls. In the window before officers arrive, bystander paralysis meant that nearby individuals who could intervene or assist often did nothing. **ComplAInSG** aims to solve this by blasting geo-targeted alerts to community members within seconds of a report, conveying passive witnesses into active first responders. Besides promoting proactivity, **ComplAInSG** helps to increase monitoring and coverage by utilising the ubiquitous smartphones as tools for assistance.

The platform serves three distinct safety scenarios: physical altercations (fights, assaults), sensitive personal crimes (molestation, harassment), and infrastructure failures (MRT faults, fire hazards). For each, it provides a reporting flow complete in under 5 seconds, AI-assisted content verification, content description generation and a structured community response layer that works in parallel with, not instead of, official emergency services.

2. The Problem - Too Little, Too Late

In Singapore, the scene of chairs being thrown, hair being pulled, and punches being thrown in hawker centres, void decks and MRT stations is a documented and recurring problem. The pattern is always the same: bystanders watch, someone eventually calls 999, and police arrive 8 to 12 minutes later. By that time, the harm is already done.

This is not a policing failure. The reality is that officers cannot be everywhere. The failure is structural. While there are dozens of people physically present at the scene, most fail to act as first responders in the critical minutes before police arrive. **ComplAInSG** aims to bridge this existing gap.

The same structural gap applies to other incidents. MRT breakdowns are reported on Reddit before SMRT makes an official announcement because commuters on the train have no fast, verified channel to report what they are witnessing. Molestation victims often

stay silent because reporting feels slow, formal, and unsafe. **ComplAInSG** addresses all of these with a single, unified platform.

3. Why ComplAIn? How We Differ from Existing Solutions

Several community safety tools exist in Singapore. We are not building a replacement. We are building the layer that is missing from all of them.

3.1 SGSecure - The Direct Comparison

SGSecure is Singapore's primary community safety app, developed by SPF. It is a strong platform for national resilience awareness and volunteer coordination. However, it was not designed for real-time, bystander-driven incident response. The key gaps **ComplAInSG** fills:

	SGSecure	***ComplAInSG***
Response model	Report to authorities; wait for official response	Immediately blasts a geo-alert to the community within 500m. Community members are first responders
Who response	Who responds to SPF and trained SGSecure volunteers. Any nearby citizen, no pre-enrolment or training required	Any nearby citizen, no pre-enrolment or training required
Reporting speed	Form-based submission requiring multiple fields	One tap to record, one tap to submit under 5 seconds. Big, persistent button on home screen.
Content verification	Human review by SPF	On-device AI immediately screens for authenticity, classifies incident type, and generates a summary before SPF sees it
Media type	Photos and text reports	Photos with AI-assisted person identification

Community coordination	Not available	Live responder map, On My Way / Recording / Calling 999 status, bystander count nudge

3.2 Why Reddit, WhatsApp, and Social Media Fall Short

Singaporeans already report incidents informally on Reddit (r/singapore), WhatsApp community groups, and Telegram channels. This shows that the activation energy exists. People want to report. But these channels have a critical flaw: they are broadcasting platforms, not response coordination tools. A post on Reddit about a fight at Bedok Interchange reaches thousands of people who are not there, and zero people who are.

ComplAInSG uses the same posting behaviour people already perform, but routes the alert only to people within 500 metres of the incident, at the moment it is happening. The activation energy is deliberately matched to social media: posting a photo or short audio clip is something every Singaporean already knows how to do. The only difference is that ComplAIn makes it faster (one tap vs. navigating to an app, typing a caption, selecting a channel) and routes it to the people who can actually help.

3.3 Advantages Over CCTV-Only Surveillance

Singapore's existing CCTV infrastructure is extensive but passive. Cameras record; they do not respond. Footage is reviewed after the fact, not during. **ComplAInSG** transforms the community's smartphones into an active, real-time monitoring and response network that works alongside CCTV:

- Every smartphone is a mobile sensor. Cameras can reach angles, indoor spaces, and blind spots that fixed CCTV cannot
- Reports are timestamped, geotagged, and AI-verified in real-time, not hours later
- Community responders are alerted while the incident is live, while CCTV footage is typically reviewed only after it is too late
- Multiple crowd-submitted images from different angles provide more complete evidence than a single fixed camera

ComplAInSG does not replace CCTV. It extends surveillance into every public space in Singapore, wherever there is a person with a phone.

4. How ComplAinSG Works

4.1 The Reporting Flow

Action	***Design Decision***
Tap Record button (1 tap)	Persistent photo button at the navigation bar and large report button at the Home Page
Select incident category (1 tap)	One tap to select. Optional.
Location Determination	GPS auto-tags simultaneously.
Tap to submit	AI screens content on-device. Report fires. Geo-blast sent. SPF notified.
AI Description	AI generates descriptions based on images. Minimal typing.

The choice of images over video is a deliberate design trade-off. Video provides richer context but introduces upload latency (especially on congested networks at incident sites), higher storage costs, slower AI processing, and privacy risks from capturing uninvolved bystanders in extended footage. Images are sufficient for incident documentation, person identification, and SPF evidence submission while enabling the sub-5-second flow that is central to the platform's safety value.

4.2 Community Alert System

When a report is verified and submitted, **ComplAInSG** pushes a geo-targeted push notification to all users within a severity-appropriate radius. The notification is direct and actionable: "Fight at Bedok Interchange, 80m from you. Can you help?" Recipients respond with a single tap: On My Way, I'm Recording, or Calling 999. A live responder map shows the original reporter that help is coming, reducing their anxiety and encouraging them to stay safe and keep documenting.

This system directly counters the bystander effect: when responsibility is explicitly assigned to an individual rather than diffused across a crowd, response rates increase significantly. The notification shows each recipient how many others have seen the alert but not yet acted as a deliberate psychological nudge grounded in behavioural science.

4.3 AI Engine

ComplAInSG's AI engine is powered by OpenAI GPT-5.1, orchestrated across three specialised agents that work in sequence the moment a report is submitted. The Text Triage Agent processes any optional text input provided by the user. A few words, a location detail, and a description of what they saw and generates a structured natural language description of the incident. That output is passed alongside the submitted image to the Vision Agent, which analyses the visual content, identifies key incident indicators such as crowd behaviour, physical posturing, and environmental context, and produces a scene description grounded in what the camera actually captured. Both outputs are then fed into the Final Triage Agent, which synthesises them into a single coherent, human-readable incident summary ready for SPF consumption. Beyond summarisation, the engine performs authenticity filtering using image metadata such as cross-referencing timestamp and device GPS data against the reported location and time to flag submissions where the data does not align, catching recycled images or fabricated reports before they ever reach the community feed or law enforcement.

5. Key Design Decisions and Trade-Offs

Decision	***Trade-off & Rationale***
Images, not video, as primary media	Video is richer but slower to upload, harder to process offline, and riskier for bystander privacy. Images are sufficient for incident documentation and person ID, and enable the sub-5-second flow. Video processing is a Phase 2 feature once the infrastructure matures.
Anonymous reporting with Singpass verification	Full anonymity enables more reporting (especially for sensitive incidents like molestation) but creates abuse risk. Singpass identity is stored on-record as a deterrent never shown publicly. This balance protects both the reporter and the platform's integrity.
Community response, not community intervention	**ComplAInSG** does not ask civilians to physically intervene in dangerous situations. Responder actions are limited to: document, call 999, be present. The platform is a coordination and documentation tool. Asking civilians to intervene

	would create liability and safety risks.
Cloud AI for accuracy	Pure cloud AI would introduce 2-4 second network latency, which may be unacceptable in a crisis. However, we chose to use it due to the accuracythat the cloud AI can provide
No live video streaming	Streaming would require sustained network bandwidth, drain battery, and create a persistent surveillance infrastructure. Discrete image/audio submissions are less invasive and more resilient to network instability.

6. How ComplAIn Can be Extended and Adapted

ComplAInSG is designed as a platform, not a standalone app. Its architecture anticipates several layers of future extension, both technical and institutional.

6.1 Video Processing and CCTV Integration

The current system uses images and audio as primary media. This is a deliberate Phase 1 constraint to ensure speed, offline reliability, and manageable infrastructure costs. In Phase 2, the AI pipeline is designed to accept short video clips (10–30 seconds) from user submissions, with on-device compression before upload. The technical pipeline is identical to the image path, but only the input format changes.

The more significant extension is CCTV integration. Singapore's public CCTV network (HDB, NParks, LTA) is extensive but operates as a closed, passive system. **ComplAInSG**'s architecture includes an API contract designed for camera feed integration: when a report is filed at a specific location, the system can request a still or short clip from the nearest public CCTV camera, feeding it directly into the person identification model. This would dramatically increase identification confidence without requiring any additional action from the user.

This integration requires a formal data-sharing agreement with HDB and LTA, and is scoped for Phase 3. The infrastructure is already designed to receive it.

6.2 Enhancing Existing Applications

ComplAInSG is not positioned to replace SGSecure, MyTransport, or 995. It is designed to complement and enhance them by filling the gap between incident detection and official response:

- SGSecure: **ComplAInSG** generates structured incident reports that can be forwarded directly to SPF via API in the same format SGSecure uses, reducing duplication and improving data quality for both platforms.
- SMRT Fault Reporting: MRT transport fault reports filed on **ComplAInSG** are automatically cross-posted to the LTA DataMall incident feed, improving the public transport information ecosystem.
- SPF I-Witness Portal: Evidence-grade media from ComplAIn (timestamped, encrypted, AI-verified) can be handed off directly to SPF's existing evidence management system. No secondary submission required from the user.

6.3 Closed Community Networks

Schools, condominiums, hospitals, and event venues can deploy private **ComplAInSG** networks scoped to their premises. A school uses it for campus safety (bullying, medical emergencies) without exposing internal incidents publicly, while serious incidents still trigger SPF notification. This B2B deployment model creates a sustainable revenue stream to fund continued development.

6.4 Global Adaptation

The core architecture is city-agnostic. Three components require localisation for a new deployment: the identity verification module (Singpass is replaced by the local digital ID equivalent, any OAuth-compatible national ID system works), and the community alert language pack. Everything else, like the AI models, the reporting flow, the evidence vault, and the community coordination layer, transfers directly.

In lower-infrastructure contexts (limited smartphone penetration, inconsistent power), the offline-first architecture means the core reporting and alerting functions remain available via Bluetooth mesh between nearby devices, with no network dependency. This makes **ComplAInSG** deployable in environments where no existing safety app functions reliably.

7. Feasibility - For the Future

7.1 Integration with Existing Applications

ComplAInSG is built to complete Singapore's existing safety infrastructure, not compete with it. When a report is authenticated, the structured brief GPS coordinates, severity classification, AI summary, and verified media are forwarded via API directly into SPF's backend in the same format SGSecure already uses. Transport fault reports cross-post into LTA's DataMall feed, enriching MyTransport.SG in real time. Medical emergency alerts are structured for compatibility with SCDF's 995 dispatch system. **ComplAInSG** generates the first sixty seconds of structured, community-sourced intelligence that existing platforms currently receive only minutes later, making every application faster and better-informed from the moment an incident occurs.

7.2 Privacy Architecture

Reporter identity is stored encrypted and linked to Singpass, never surfaced to other users or shown in any community view. All suspect imagery is algorithmically blurred in public feeds; unblurred access is restricted to verified SPF officers. Location data is not retained beyond 72 hours. All data collection requires explicit informed consent and is deletable on user request.

7.3 Offline and Low-Connectivity Operation

ComplAIn degrades gracefully rather than failing in constrained environments. On full network: all features available. On limited connectivity (2G or congested): reports queue locally, Bluetooth mesh alerting activates for nearby users, and upload resumes on reconnect. On no network: recording continues, GPS caches location, Bluetooth/WiFi Direct alerts reach peers within 100m. On low battery (<15%): background AI suspends, core one-tap reporting remains available. The system is designed for Singapore's dense urban environment but is robust enough to function at outdoor events, in basement carparks, and in MRT tunnels.

8. Limitations

- Cold start: the alert network has limited value until a critical mass is reached. We estimate 2% Singapore smartphone penetration (~100,000 users) as the threshold where virtually every residential and commercial area has at least 3–5 **ComplAInSG** users within any 250m radius. Early adoption strategy requires partnerships with community groups, schools, and grassroots organisations.
- Alert fatigue: high alert volume risks users disabling notifications. Aggressive geofencing, user-configurable quiet hours, and severity thresholds are essential mitigations requiring continuous calibration.
- AI model limitations at night / in rain: the current person identification model is trained primarily on daytime, clear-weather footage. Nighttime-augmented training data is a known gap in the Phase 1 model.
- Video processing deferred: full video analysis is a Phase 2 feature. Phase 1 relies on images and audio, which cover the majority of incident documentation needs but may miss the temporal context available only in video.

9. Conclusion

When a chair is thrown at a hawker centre, the people who can stop the harm are already there. They just lack the tools. **ComplAInSG** gives them a five-second, cost-free, offline way to help without asking them to be heroes only to be witnesses. By streamlining the organic reporting Singaporeans already do on Reddit and WhatsApp, we aren't replacing the SPF or SGSecure. We are bridging the critical 8-minute gap between an incident and the official response. It’s about filling that silence in every neighbourhood, every day.